

Early game indicators in professional league of legends

- WHICH EARLY GAME STATS ARE MOST LINKED TO WINNING?
- CAN WE PREDICT THE WINNER USING ONLY EARLY-GAME DATA?
- DOES IMPORTANCE OF EARLY GAME CHANGE ACROSS SEASONS?
- DO DIFFERENT REGIONS SHOW DIFFERENT PATTERNS?

Data sources

- Dataset: Oracle's elixir , (season 2019,2022,2025)
- Regions (LCK, LPL,LEC,LCS,PCS)
- Key columns : gold – xp – cs ,diff at 15 minutes, first objectives ,kills , game length
- Preprocessing : merged 3 files , dropped nulls , created new columns(gold_lead,got_inhibitor)

Key results until now

OBJECTIVE WIN RATE RANKING:

- Inhibitor 93%
- First baron:85%
- First to 3 turrets : 78%
- First blood 60%
- First dragon 58%

EARLY COMBINED INDICATORS

0/5 lead to 13.5%win rate

1/5 lead to 24.2% win rate

2/5 lead to 38.8% win rate

3/5 lead to 61.3% win rate

4/5 lead to 78% win rate

5/5 lead to 86.5% win rate

- (first blood , gold lead at 15 minutes , first tower , first herald , first dragon)

Baron is the biggest factor

- Teams ahead with baron kill : 91.9%winrate
- Teams ahead without baron kill : 44.5%winrate
- Teams behind with baron kill:70.7% winrate (comeback tool).

The next steps

- More detailed data analysis and findings of correlations
- Knime visualizations : heatmaps , bar charts , trend lines
- Notebook narrative
- Github repo + youtube video